

determining the requirements, the planning for the software development is done in the second stage, with software engineers and programmers communicating with each other. This stage includes creating flowcharts, documentation and planning the timeline for the software development.

After everything is planned, the software engineers assigned for the third stage (modelling) of the project start working. At this stage, a basic working prototype of the planned software is created. Then comes the fourth stage, which is called construction, and this is when the actual software is designed and coded. The software testing is also done at this stage by software testers. On successful testing, the software is taken to the fifth and the final stage, which is deployment – where it is released to the customer or end user for use. After the user runs the software, the feedback is collected. If any error is found, then the software development process goes to another iteration of the same development cycle.

The use of AI models helps develop an effective AI system, with collaboration not only between humans, but also with machines. In this model, various machines are connected with each other using a cloud based network, and the actual data is stored on the cloud and securely exchanged between nodes. With this model, the connected AI machines can communicate with each other and develop the learning process. Humans can also develop the algorithms stored on the cloud network by following the collaborative model of software development.

FOSS tools for AI

There are various FOSS tools available for the development of AI systems. TensorFlow, Keras, Caffe, OpenNN, Theano, Mycroft and TensorRT are some examples of them. Among these tools, TensorFlow and TensorRT are best known for their use in the



Figure 1: The collaborative model of software development

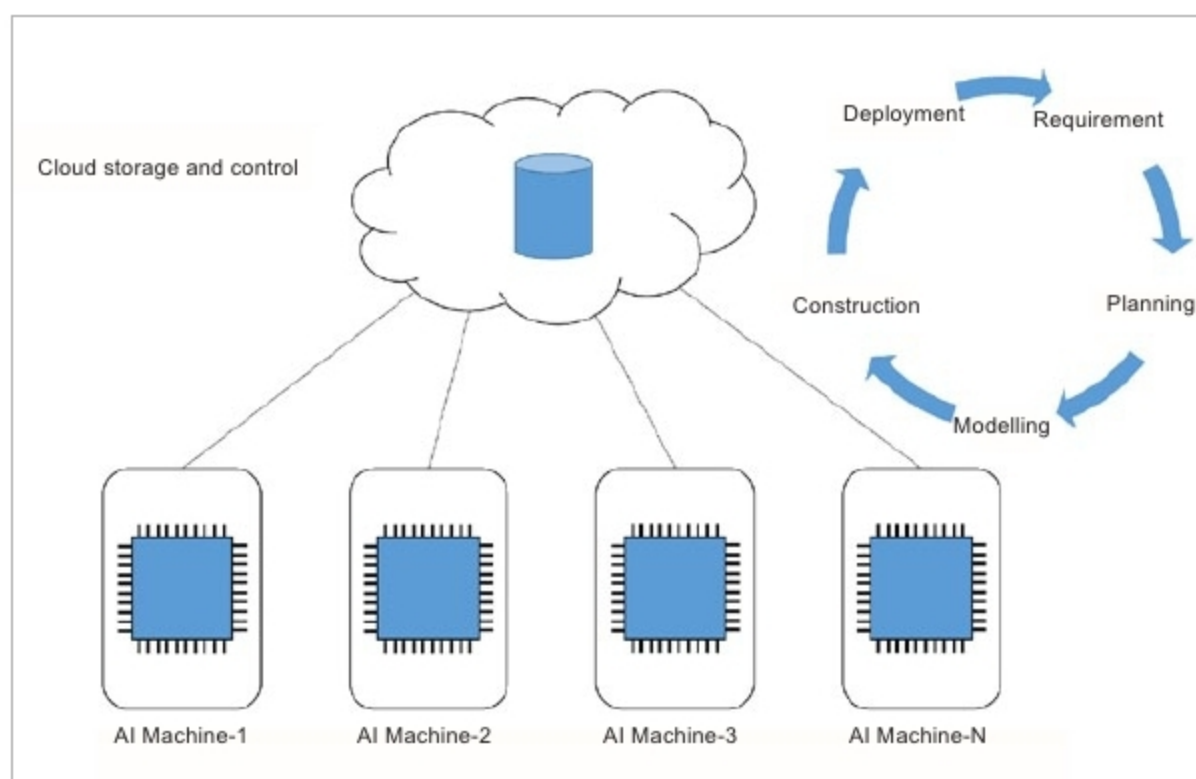


Figure 2: Collaborative AI

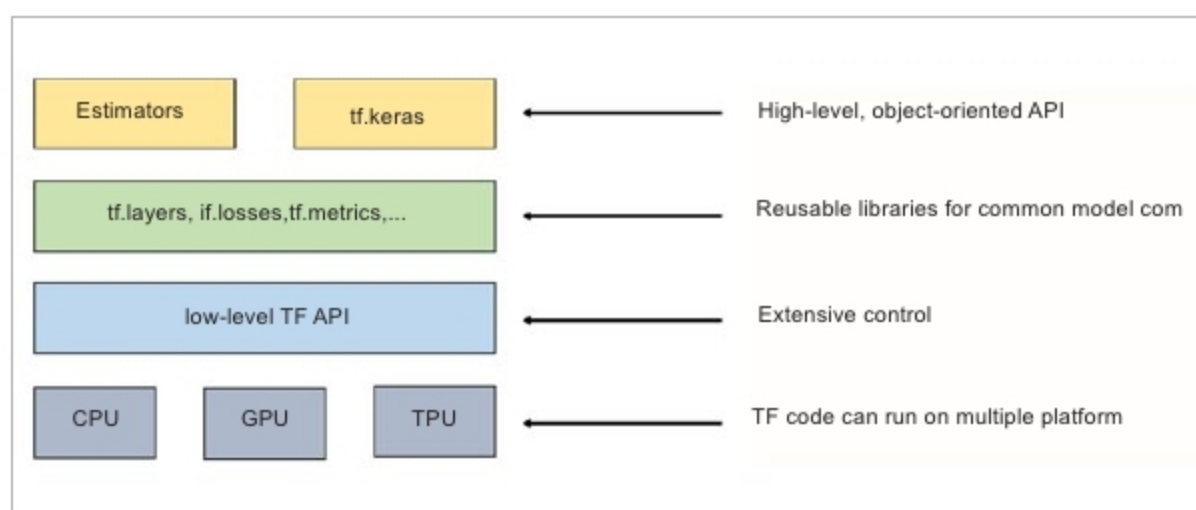


Figure 3: TensorFlow toolkit hierarchy